

Supplemental Material for: Non-additive Strain–Dopant Energetics in Quasi-Hexagonal C₆₀ Graphullerene

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I. FINITE-DIFFERENCE DERIVATION OF THE FOUR-CORNER COUPLING ESTIMATOR

II. DISCRETE STRAIN–SUBSTITUTION INTERACTION

The production estimator is a discrete interaction at fixed dopant species. B, N, and P are chemical labels, not three numerical samples of one continuous coordinate δ . For a fixed species δ and a biaxial load ϵ ,

$$\mathcal{S}_\delta(\epsilon) \equiv \frac{E(\epsilon, \delta) - E(\epsilon, 0) - E(0, \delta) + E(0, 0)}{N_{\text{atoms}}}, \quad (1)$$

which is the same four-corner combination as Eqs. (2) and (3). It cancels the strain-only and substitution-only increments and retains their joint remainder. It is not $\partial^2 E / \partial \epsilon \partial \delta$.

A mixed derivative appears only after a separate, local composition coordinate x is introduced at that fixed species. In that local representation the cross coefficient below is a finite-difference image of $\partial^2 e / \partial \epsilon \partial x$, not a derivative along a shared B/N/P axis.

A. Local expansion at fixed species

Let $e(\epsilon, x)$ be the total energy per atom, with ϵ the biaxial load and x a local composition coordinate defined only after the species δ has been fixed. About the pristine, unloaded reference $(\epsilon, x) = (0, 0)$,

$$e(\epsilon, x) = e_0 + e_\epsilon \epsilon + e_x x + \frac{1}{2} e_{\epsilon\epsilon} \epsilon^2 + \frac{1}{2} e_{xx} x^2 + e_{\epsilon x} \epsilon x + \mathcal{O}(3), \quad (2)$$

where subscripts denote partial derivatives evaluated at the reference and $e_{\epsilon x} \equiv \partial^2 e / \partial \epsilon \partial x$. The four corners of one species are $e(0, 0)$, $e(\epsilon, 0)$, $e(0, x)$, and $e(\epsilon, x)$. Forming the inclusion–exclusion combination gives

$$\begin{aligned} e(\epsilon, x) - e(\epsilon, 0) - e(0, x) + e(0, 0) \\ = e_{\epsilon x} \epsilon x + \mathcal{O}(\epsilon^2 x, \epsilon x^2, \epsilon^3, x^3). \end{aligned} \quad (3)$$

Every term that depends on ϵ alone or on x alone cancels. Dividing by ϵx isolates the mixed derivative of this local chart,

$$\frac{e(\epsilon, x) - e(\epsilon, 0) - e(0, x) + e(0, 0)}{\epsilon x} = e_{\epsilon x} + \mathcal{O}(\epsilon, x). \quad (4)$$

The main-text order parameter is this interaction reported per atom at the production load (+3%) and at

the discrete label δ , without dividing by a continuous Δx . The same cancellation is what Eq. (3) states for total energies.

B. Why the four-corner quotient is not the local slope

Dividing the per-atom interaction by the load,

$$\frac{\mathcal{S}_\delta(\epsilon)}{\epsilon} = e_{\epsilon x} x + \mathcal{O}(\epsilon, x), \quad (5)$$

still carries the local composition step x . It is therefore not equal to $e_{\epsilon x}$, and it is not a derivative with respect to dopant identity. Equation (4) uses that quotient only as a finite-difference image inside one local chart. The manuscript reports $\mathcal{S}_\delta(\epsilon)$ itself.

C. What the four corners remove

A strain-only scan at fixed $\delta = 0$ retains $e_\epsilon \epsilon + \frac{1}{2} e_{\epsilon\epsilon} \epsilon^2$. A substitution-only scan at fixed $\epsilon = 0$ retains $e_x x + \frac{1}{2} e_{xx} x^2$. Adding those two scans double-counts nothing in the cross term and still omits $e_{\epsilon x} \epsilon x$. The four-corner combination deletes both one-variable polynomials through the order kept in Eq. (2) and retains the mixed term. That is the sense in which sequential scans are additive and the joint evaluation is not: the omitted piece is $e_{\epsilon x} \epsilon x$, up to higher-order remainders.

D. Gauge and higher-order content

Shifting the energy zero cancels inside Eq. (3), so $\mathcal{S}_\delta$ does not depend on a global energy offset. At the production load of +3% and at a full substitution on the doped cell, the steps in ϵ and in local x are not infinitesimal. Cubic and higher terms then sit inside the reported interaction. The manuscript does not separate them from $e_{\epsilon x} \epsilon x$. Table II records that limit in one line: the mixed derivative is defined only inside the local chart, and the higher-order remainder is not resolved.

E. Propagation of corner noise

For four uncorrelated corner energies with equal standard uncertainty σ_E , the numerator of Eq. (1) has un-

TABLE I. Positioning relative to representative prior work on qHP graphullerene and related covalent networks (not an exhaustive literature map). “Joint (ϵ, δ) ” denotes simultaneous strain and substitutional doping in the same calculation series; “four-corner total-energy audit” denotes symmetric (ϵ, δ) corner differences as in the main-text definition of $\mathcal{S}$.

Prior focus	Joint (ϵ, δ)
Separate strain or doping maps [1–7]	×
Nonlinear band-edge / transport response [8–10]	Partial
Alloy cluster expansion (effective $c\epsilon x$) [11, 12]	✓
This work (qHP C ₆₀ , PBE+D3)	✓

certainty $2\sigma_E$. Per atom, with $\sigma_e = \sigma_E/N_{\text{atoms}}$,

$$\sigma_S = 2\sigma_e. \quad (6)$$

This is the same quadrature as Eq. (5). Correlated SCF noise across chemically related corners would reduce the effective coefficient below 2; the main-text reporting band uses $2\sigma_e$ as a conservative reporting floor (Table I). That floor is a statistical envelope on corner noise. It is distinct from the rigid-protocol estimate, which compares the fixed-coordinate evaluation with the fully relaxed fixed-cell evaluation studied in the main text and is not a bound on every constraint manifold.

III. LITERATURE POSITIONING AND DESCRIPTOR COMPARISON

IV. ARCHIVED LEGACY-PBE TETRAMER BASELINE

Table III reproduces the archived tetramer cluster map (PBE without D3; Cartesian coordinates not affinely scaled) retained for reproducibility; it is *not* used for quantitative strain-physics claims in the main text.

V. CALCULATION INVENTORY (REPRODUCIBILITY)

Tables IV and V list the DFT groups underlying the main-text claims and superseded exploratory grids retained only for reproducibility. They are *not* scientific results tables.

Four-corner E	Nonlinear audit
×	×
×	×
×	×
✓	✓

VI. FULL TETRAMER π -DOS AT $\epsilon=0$

VII. PERIODIC FRONTIER-ORBITAL LOCALIZATION

Cage-resolved VBM/CBM isosurfaces on periodic $n \times \text{C}_{60}$ cells ($n = 2\text{--}8$) at $\epsilon=0$ complement the tetramer π -DOS above and the main-text electronic taxonomy (donor/acceptor vs. size dilution). Carbon cages are shown as uniform gray MSMS molecular surfaces (probe 1.3 Å); heteroatoms use element colors.

VIII. PERIODIC COUPLING-ENERGY ROWS ($n \geq 6$)

IX. FOUR-CORNER TOTAL ENERGIES (+3%)

X. MARCUS REORGANIZATION ENERGIES

Polaron transport diagnostics are reported here for completeness; the main-text analysis centers on $\mathcal{S}$.

XI. HOPPING INTEGRAL J

On the tetramer dimer grid, B substitution at $\epsilon=0$ delocalizes the charged-state orbital (IPR 75.0 $\rightarrow$ 24.6) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives $J = 37.2$ meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [4] is *not* met at PBE+D3 ($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

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- [1] R. Wang, H. Li, M. A. Shakoory, X. Cheng, Y. Hu, and L. Wang, Strain engineering of electronic structure and thermoelectric properties of quasi-hexagonal fullerene monolayer, *J. Appl. Phys.* **136**, 015101 (2024).
[2] M. A. Khan, M. Atif, and M. N. Leuenberger, Tuning the electronic and optical properties of impurity-engineered two-dimensional graphullerene half-semiconductors, *Phys. Rev. Mater.* **9**, 034001 (2025).
[3] B. Peng and M. Pizzochero, Monolayer C₆₀ networks: a first-principles perspective, *Chem. Commun.* **61**, 10287 (2025).
[4] A. Capobianco, J. Wiktor, A. Landi, F. Ambrosio, and A. Peluso, Electron localization and mobility in mono-

- layer fullerene networks, *Nano Lett.* **24**, 8335 (2024).
[5] R. M. Tromer, L. A. Ribeiro, Jr., and D. S. Galvão, A DFT study of the electronic, optical, and mechanical properties of a recently synthesized monolayer fullerene network, *Chem. Phys. Lett.* **804**, 139925 (2022).
[6] G. Y. Li, H. Wang, and M. Zhang, Two-dimensional graphullerene C₂₄: Strain-induced tunable electronic and optical properties for applications in wearable UV-protective devices, *ACS Applied Nano Materials* **7**, 23145 (2024).
[7] J. Wang, K. Li, and H. Zhang, simulation on stress-related anisotropy of qTP C₆₀ and qHP C₆₀: implications for optoelectronic nanodevices, *ACS Applied Nano*

TABLE II. Positioning of the strain-dopant coupling energy $\mathcal{S}$ (main-text four-corner estimator) against common stability and electronic descriptors. Algebraically $\mathcal{S}$ has the form of an interaction (coupling) energy; the distinction is *dimensional*, i.e. which axes of the joint (ϵ, δ) control space each quantity resolves. “Additive-screening error” denotes the bias incurred when total energies under joint load are assembled additively from separate strain-only and doping-only scans; $\mathcal{S}$ equals that bias by construction. “Double-reference cross term” denotes a symmetric difference of differences over the four (ϵ, δ) corners rather than subtraction of a single reference state or fragment pair. Entries are conceptual scope, not numerical performance; consistent with its energetic definition, $|\mathcal{S}|$ is *not* reducible to a single geometric ($\Delta\bar{d}$) or charge (Δq) channel alone.

Descriptor	Nonlinear (ϵ, δ) coupling	Additive-screening error	Double-reference cross term	Joint strain + doping design
Formation / substitution energy E_{sub}	×	×	×	×
Binding / interaction energy ΔE_{int}	Partial	×	×	Partial
Linear strain coefficient α	×	×	×	Partial
Band gap E_g / work function	×	×	×	×
$\mathcal{S}$ (this work)	✓	✓	✓	✓

TABLE III. Archived legacy-PBE (no D3) tetramer cluster map retained for reproducibility only; not used for main-text quantitative claims. Cartesian coordinates are not affinely scaled. $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ (no chemical potentials; relative ranking within this protocol only).

System	n_{dop}	$E(\epsilon=0)$ (E_h)	$E(\epsilon=+5\%)$ (E_h)	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	−1363.5512	−1363.5509	−1.0	—
B-doped	13	−1328.4900	−1328.4090	+56.9	+73.4
N-doped	12	−1414.3812	−1414.4053	−303.2	−115.3
P-doped	12	−1370.4965	−1370.4888	−19.1	−15.8

Materials **7**, 3456 (2024).

- [8] F. Lee, M. Tripathi, R. S. Salas, S. P. Ogilvie, A. A. Graf, I. Jurewicz, and A. B. Dalton, Localised strain and doping of 2D materials, *Nanoscale* **15**, 7227 (2023).
- [9] T. S. A. Cassiano, M. L. Pereira, Jr., G. M. e Silva, P. H. de Oliveira Neto, and L. A. Ribeiro, Jr., Large polarons in two-dimensional fullerene networks: the crucial role of anisotropy in charge transport, *Nanoscale* **16**, 2337 (2024).
- [10] P. Santra, S. Ghaderzadeh, M. Ghorbani-Asl, H.-P. Komsa, E. Besley, and A. V. Krashenninnikov, Strain-modulated defect engineering of two-dimensional materials, *npj 2D Mater. Appl.* **8**, 33 (2024).
- [11] J. M. Sanchez, F. Ducastelle, and D. Gratias, Generalized cluster description of multicomponent systems, *Physica A* **128**, 334 (1984).
- [12] D. B. Laks, L. G. Ferreira, S. Froyen, and A. Zunger, Efficient cluster expansion for substitutional systems, *Phys. Rev. B* **46**, 12587 (1992).

TABLE IV. DFT calculation groups supporting the main-text claims (reproducibility inventory; Supplemental Material only). Superseded exploratory grids appear in Table V.

Calculation group	System	Count	Protocol
Archived baseline (legacy PBE)	$4 \times \text{C}_{60}$	24	Rigid strain; not used for main-text claims
π -DOS and gaps	$4 \times \text{C}_{60}$	12	PBE+D3
IPR (polaron dimer)	Dimer	4	Pristine/B; $\epsilon=0, +3\%$
Hopping J	Dimer	4	PBE+D3
Charged polarons ^a	$4 \times \text{C}_{60}$	20	UKS; geometry opt. + vertical SP
Periodic size scaling	$n \times \text{C}_{60}$	41	PBE+D3; 400/350 Ry
Cutoff benchmark	$6 \times \text{C}_{60}$ N	1	+3%; 400 Ry single-point
Ionic relaxation	$4 \times \text{C}_{60}$	4	Fixed-cell GEO; P corners
Ionic relaxation	$1 \times \text{C}_{60}$ P	4	Fixed-cell GEO; periodic
Alternate placement	$4 \times \text{C}_{60}$	24	Rigid PBE+D3
Periodic placement	$4 \times \text{C}_{60}$	16	$n=4$; two independent maps
Matched-functional rigid	$4 \times \text{C}_{60}$	4	PBE+D3 at relaxed coordinates
Reference placement	$4 \times \text{C}_{60}$	24	Rigid PBE+D3
Hirshfeld population	$1 \times \text{C}_{60}$ B/N/P	18	Spin-polarized KS

^aSix relaxed tetramer geometries provide charged-state initial structures.

TABLE V. Exploratory DFT grids superseded by the production protocol (reproducibility archive only).

Task group	System	N_{conv}	Disposition
Early strain probe	$2 \times \text{C}_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2 \times \text{C}_{60}$	24	Superseded
Exploratory design grid	$4 \times \text{C}_{60}$	20	Archive only

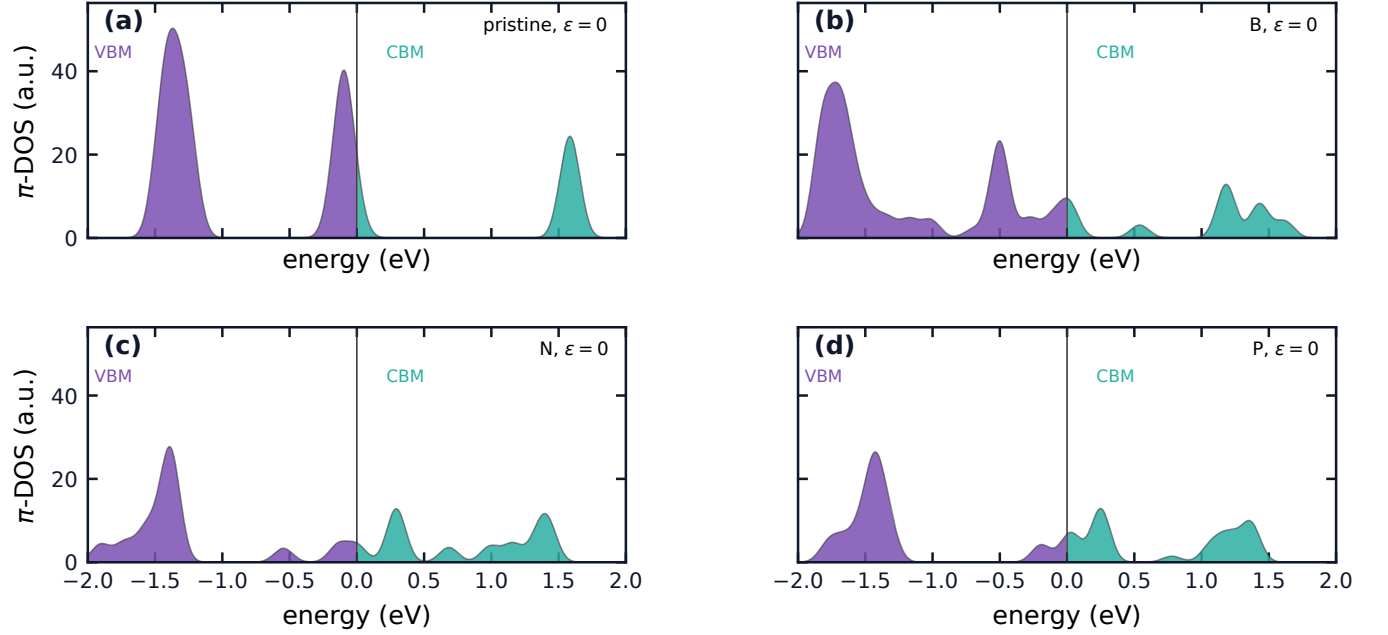
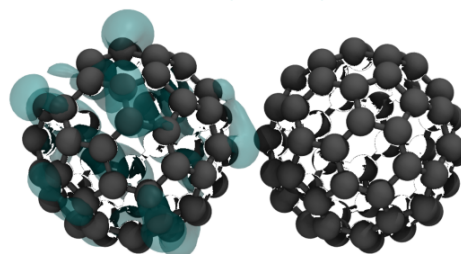


FIG. 1. Full π -DOS panels for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3), complementary to the main-text electronic-structure figure.

qHP C_{60} graphullerene ($\epsilon = 0\%$, N-doped)

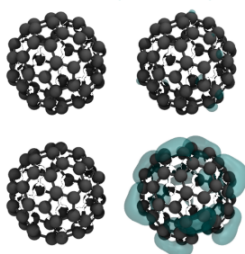
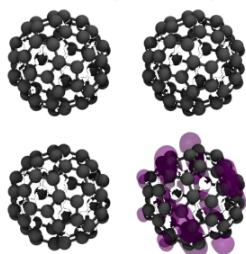
$n = 2$ ($2 \times C_{60}$, N-doped)
VBM (HOMO)

$n = 2$ ($2 \times C_{60}$, N-doped)
CBM (LUMO)



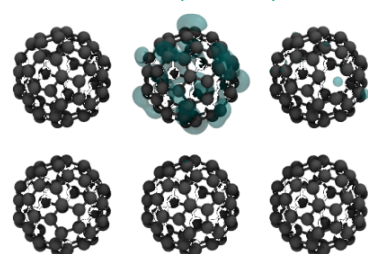
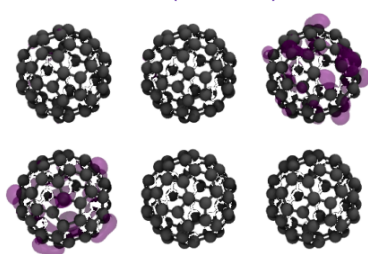
$n = 4$ ($4 \times C_{60}$, N-doped)
VBM (HOMO)

$n = 4$ ($4 \times C_{60}$, N-doped)
CBM (LUMO)



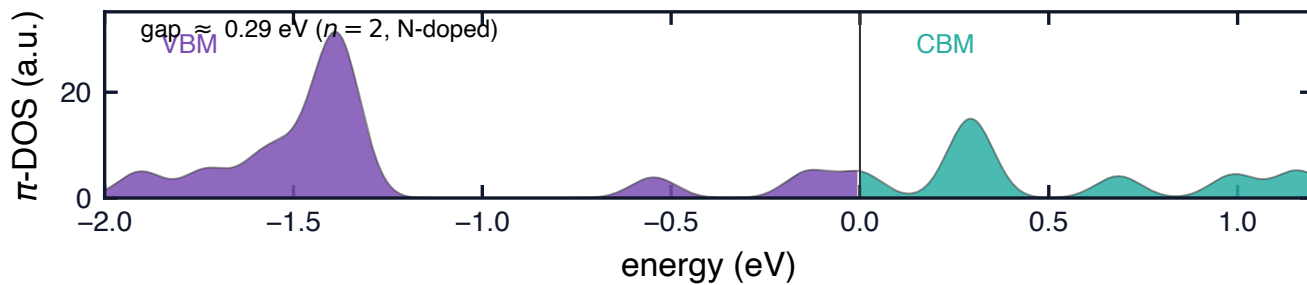
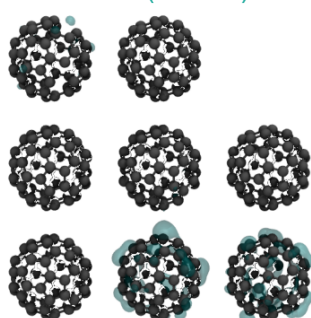
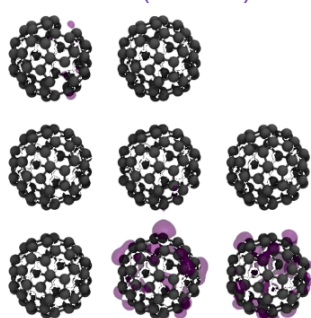
$n = 6$ ($6 \times C_{60}$, N-doped)
VBM (HOMO)

$n = 6$ ($6 \times C_{60}$, N-doped)
CBM (LUMO)



$n = 8$ ($8 \times C_{60}$, N-doped)
VBM (HOMO)

$n = 8$ ($8 \times C_{60}$, N-doped)
CBM (LUMO)



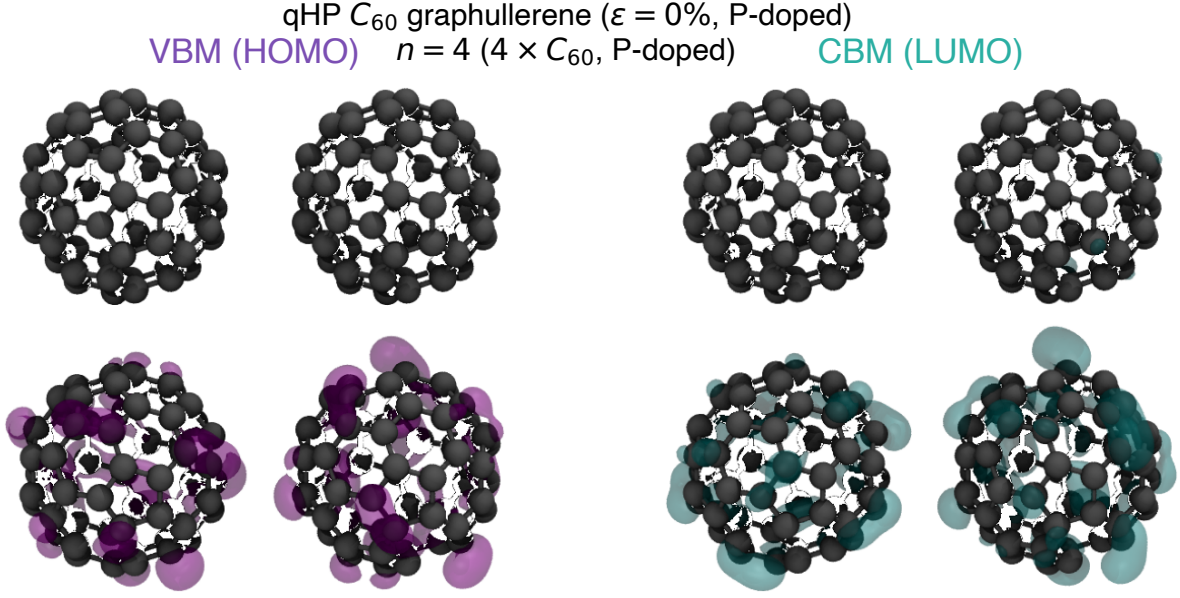


FIG. 3. P-doped $n=4$ supercell at $\epsilon=0$: VBM (left) and CBM (right) on gray MSMS cages. Both frontier states localize on a subset of cages, providing a spatial counterpart to the large periodic $|S|$ of size-mismatched P reported in the main text.

TABLE VI. Supplemental periodic coupling-energy rows at +3% ($n \geq 6$; 350 Ry production cutoff; completeness only).

n	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
6	B	+3	-7.76
6	N	+3	+2.38
6	P	+3	-31.89
8	B	+3	-11.26
8	N	+3	-2.30
8	P	+3	-29.94

TABLE VII. Machine-readable four-corner total energies per atom (Ha/atom) underlying $\mathcal{S}$ at +3% biaxial strain (PBE+D3, rigid protocol, reference placement seed 42). Corners follow Eq. (3): $E_{00} = E(0,0)$, $E_{e0} = E(\Delta\epsilon,0)$, $E_{0\delta} = E(0,\Delta\delta)$, $E_{e\delta} = E(\Delta\epsilon,\Delta\delta)$, all per atom relative to the same supercell size $n \times C_{60}$. $\mathcal{S}$ (meV/atom) matches Eq. (2) and the open audit chain (CSV mirror in the public data release).

n	δ	ϵ (%)	E_{00}	E_{e0}	$E_{0\delta}$	$E_{e\delta}$	$\mathcal{S}$
1	B	+3	-5.6826713370	-5.6825614149	-5.6339765778	-5.6341744697	-8.38
1	N	+3	-5.6826713370	-5.6825614149	-5.7532834847	-5.7530890722	+2.30
1	P	+3	-5.6826713370	-5.6825614149	-5.6922302079	-5.6932928189	-31.91
2	B	+3	-5.6817028549	-5.6817462745	-5.6338466634	-5.6339702803	-2.18
2	N	+3	-5.6817028549	-5.6817462745	-5.7532007583	-5.7529924491	+6.85
2	P	+3	-5.6817028549	-5.6817462745	-5.6895002016	-5.6905831259	-28.29
4	B	+3	-5.6828203886	-5.6827081746	-5.6341906516	-5.6343073761	-6.23
4	N	+3	-5.6828203886	-5.6827081746	-5.7535095848	-5.7532889091	+2.95
4	P	+3	-5.6828203886	-5.6827081746	-5.6924497414	-5.6932089910	-23.71
6	B	+3	-5.6828517956	-5.6827379495	-5.6340195050	-5.6341908867	-7.76
6	N	+3	-5.6828517956	-5.6827379495	-5.7533481235	-5.7531468231	+2.38
6	P	+3	-5.6828517956	-5.6827379495	-5.6916511984	-5.6927091482	-31.89
8	B	+3	-5.6828645675	-5.6825524079	-5.6341364256	-5.6342380770	-11.26
8	N	+3	-5.6828645675	-5.6825524079	-5.7534470201	-5.7532194756	-2.30
8	P	+3	-5.6828645675	-5.6825524079	-5.6919097127	-5.6926978180	-29.94

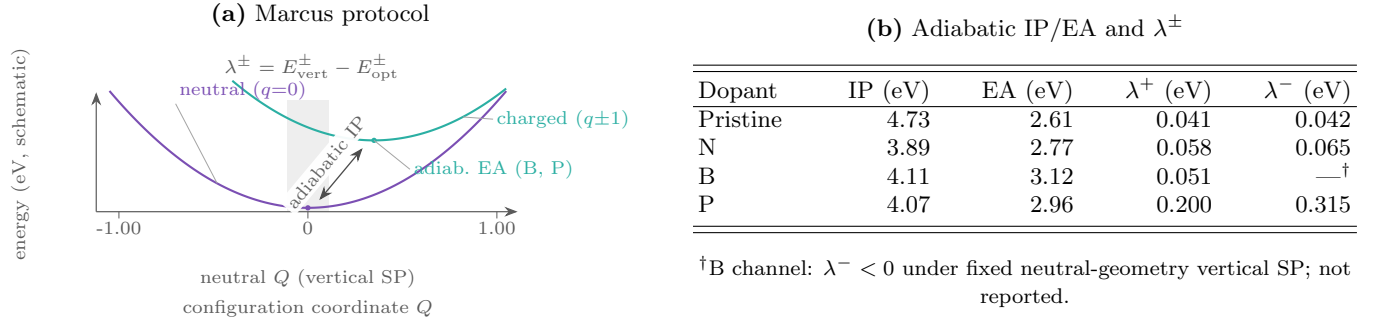


FIG. 4. Marcus reorganization energies from charged-state geometry optimizations and vertical single-points at the neutral geometry. Panel (a) schematically defines the vertical–adiabatic decomposition following Ref. 4; panel (b) tabulates adiabatic IP/EA and $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ for pristine and B/N/P tetramers. The electron-channel λ^- for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

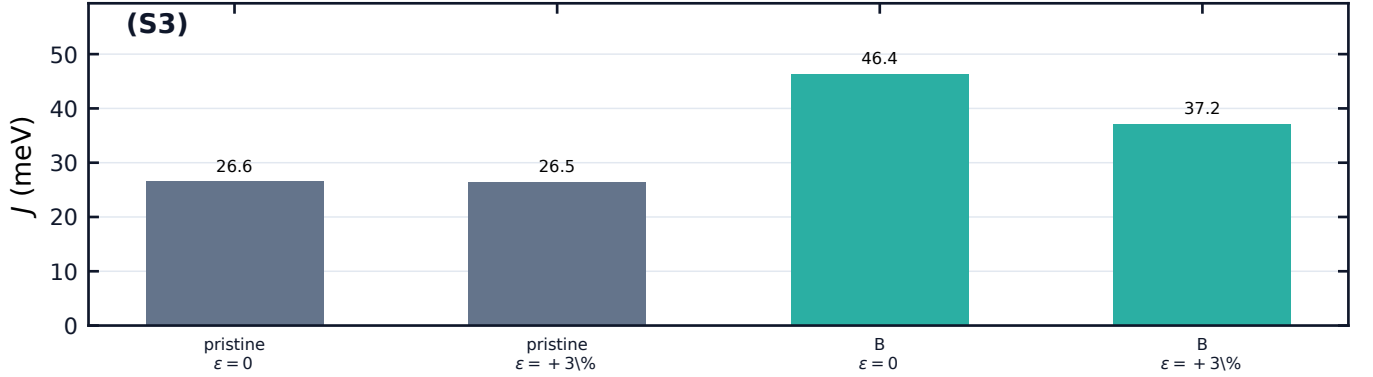


FIG. 5. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at $\epsilon=0$ and $+3\%$; PBE+D3). Companion inverse-participation-ratio values for the same four corners are tabulated in the open data release (Methods).